



INFORME DE CALIBRACIÓN (CALIBRATION REPORT)

INFORMACIÓN GENERAL (General information)

LABORATORIO (Laboratory): INGENCA SRL
DIRECCIÓN LABORATORIO (Laboratory address): Ana Monterroso de Lavalleya 2035 of. 502
SITIO DE CALIBRACIÓN (Calibration site): Laboratorio INGENCA
CLIENTE (Customer): OMEGA
DATOS DE CONTACTO (Contact information): Uruguay 1283, Montevideo.
PROCEDIMIENTO UTILIZADO (Procedure): ITT 01
N.º CERTIFICADO (Certificate N°): 7213A

INFORMACIÓN DE CALIBRACIÓN/ENSAYO (Calibration / Test Information)

EQUIPO (Instrument): Valija de inyección
MARCA (Manufacturer): Vebko
MODELO (Model): AMT105
N/S (Serial number): 4030711083
CÓDIGO (Code): -
FECHA DE EMISIÓN DE INFORME (Report date): 23/12/2025
FECHA CALIBRACIÓN/ENSAYO (Calibration/Test date): 6/11/2025
PRÓXIMA CALIBRACIÓN SUGERIDA (Next recommended calibration): 6/11/2026
MAGNITUDES/RANGOS CALIBRADOS (Physical quantities / Calibrated ranges): Tensión DC/210 V
 Tensión AC @50Hz/150 V
 Corriente AC @50Hz/32 A
 Corriente DC/32 A

INSTRUMENTAL DE REFERENCIA (Reference standards)

Nº INVENTARIO (Asset code)	EQUIPO (Instrument)	N.º Certificado (Certificate N°)	Laboratorio (Laboratory)
PATR-001	Multímetro de precisión	241530	UTE

CONDICIONES AMBIENTALES (Environmental conditions)

TEMPERATURA (Temperature)	Hum. RELATIVA (Relative humidity)	EQUIPO (Instrument)
22,4 °C	54%	ECAL-083

**DIRECTOR
(Director)**

Germán Bardier

ETIQUETA (Label)

CALIBRADO

Rev. For.	Detalle	Fecha	Resp.
5	Cambios generales.	17/10/2025	GBV

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Tensión DC V1 (DC Voltage V1)				
#	RANGO EBC (IUC RANGE)	VALOR APLICADO (APPLIED VALUE)	DESVÍO (DEVIATION)	INCERTIDUMBRE (UNCERTAINTY)
1	210,00 V	200,028 V	-0,03 V	5,6E-02 V

Tensión DC V2 (DC Voltage V2)				
#	RANGO EBC (IUC RANGE)	VALOR APLICADO (APPLIED VALUE)	DESVÍO (DEVIATION)	INCERTIDUMBRE (UNCERTAINTY)
1	210,00 V	199,997 V	0,00 V	5,6E-02 V

Tensión DC V3 (DC Voltage V3)				
#	RANGO EBC (IUC RANGE)	VALOR APLICADO (APPLIED VALUE)	DESVÍO (DEVIATION)	INCERTIDUMBRE (UNCERTAINTY)
1	210,00 V	200,011 V	-0,01 V	5,6E-02 V

Tensión AC @50Hz V1 (AC @50Hz Voltage V1)				
#	RANGO EBC (IUC RANGE)	VALOR APLICADO (APPLIED VALUE)	DESVÍO (DEVIATION)	INCERTIDUMBRE (UNCERTAINTY)
1	150,00 V	150,001 V	0,00 V	8,9E-01 V

Tensión AC @50Hz V2 (AC @50Hz Voltage V2)				
#	RANGO EBC (IUC RANGE)	VALOR APLICADO (APPLIED VALUE)	DESVÍO (DEVIATION)	INCERTIDUMBRE (UNCERTAINTY)
1	150,00 V	149,997 V	0,00 V	8,9E-01 V

Tensión AC @50Hz V3 (AC @50Hz Voltage V3)				
#	RANGO EBC (IUC RANGE)	VALOR APLICADO (APPLIED VALUE)	DESVÍO (DEVIATION)	INCERTIDUMBRE (UNCERTAINTY)
1	150,00 V	150,004 V	0,00 V	8,9E-01 V

Corriente AC @50Hz I1 OUTPUT A (AC @50Hz Current I1 OUTPUT A)				
#	RANGO EBC (IUC RANGE)	VALOR APLICADO (APPLIED VALUE)	DESVÍO (DEVIATION)	INCERTIDUMBRE (UNCERTAINTY)
1	32,000 A	9,97732 A	0,023 A	2,3E-02 A

Corriente AC @50Hz I2 OUTPUT A (AC @50Hz Current I2 OUTPUT A)				
#	RANGO EBC (IUC RANGE)	VALOR APLICADO (APPLIED VALUE)	DESVÍO (DEVIATION)	INCERTIDUMBRE (UNCERTAINTY)
1	32,000 A	9,97377 A	0,026 A	2,3E-02 A

Corriente AC @50Hz I3 OUTPUT A (AC @50Hz Current I3 OUTPUT A)				
#	RANGO EBC (IUC RANGE)	VALOR APLICADO (APPLIED VALUE)	DESVÍO (DEVIATION)	INCERTIDUMBRE (UNCERTAINTY)
1	32,000 A	9,96925 A	0,031 A	2,8E-02 A

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Corriente AC @50Hz I1 OUTPUT B (AC @50Hz Current I1 OUTPUT B)

#	RANGO EBC (IUC RANGE)	VALOR APLICADO (APPLIED VALUE)	DESVÍO (DEVIATION)	INCERTIDUMBRE (UNCERTAINTY)
1	32,000 A	9,97801 A	0,022 A	2,3E-02 A

Corriente AC @50Hz I2 OUTPUT B (AC @50Hz Current I2 OUTPUT B)

#	RANGO EBC (IUC RANGE)	VALOR APLICADO (APPLIED VALUE)	DESVÍO (DEVIATION)	INCERTIDUMBRE (UNCERTAINTY)
1	32,000 A	9,97671 A	0,023 A	2,3E-02 A

Corriente AC @50Hz I3 OUTPUT B (AC @50Hz Current I3 OUTPUT B)

#	RANGO EBC (IUC RANGE)	VALOR APLICADO (APPLIED VALUE)	DESVÍO (DEVIATION)	INCERTIDUMBRE (UNCERTAINTY)
1	32,000 A	9,97666 A	0,023 A	2,3E-02 A

Corriente DC I1 OUTPUT A (DC Current I1 OUTPUT A)

#	RANGO EBC (IUC RANGE)	VALOR APLICADO (APPLIED VALUE)	DESVÍO (DEVIATION)	INCERTIDUMBRE (UNCERTAINTY)
1	32,000 A	9,98275 A	0,017 A	1,7E-02 A

Corriente DC I2 OUTPUT A (DC Current I2 OUTPUT A)

#	RANGO EBC (IUC RANGE)	VALOR APLICADO (APPLIED VALUE)	DESVÍO (DEVIATION)	INCERTIDUMBRE (UNCERTAINTY)
1	32,000 A	9,97943 A	0,021 A	1,7E-02 A

Corriente DC I3 OUTPUT A (DC Current I3 OUTPUT A)

#	RANGO EBC (IUC RANGE)	VALOR APLICADO (APPLIED VALUE)	DESVÍO (DEVIATION)	INCERTIDUMBRE (UNCERTAINTY)
1	32,000 A	9,98476 A	0,015 A	1,7E-02 A

Corriente DC I1 OUTPUT B (DC Current I1 OUTPUT B)

#	RANGO EBC (IUC RANGE)	VALOR APLICADO (APPLIED VALUE)	DESVÍO (DEVIATION)	INCERTIDUMBRE (UNCERTAINTY)
1	32,000 A	9,98020 A	0,020 A	1,7E-02 A

Corriente DC I2 OUTPUT B (DC Current I2 OUTPUT B)

#	RANGO EBC (IUC RANGE)	VALOR APLICADO (APPLIED VALUE)	DESVÍO (DEVIATION)	INCERTIDUMBRE (UNCERTAINTY)
1	32,000 A	9,98058 A	0,190 A	1,7E-02 A

Corriente DC I3 OUTPUT B (DC Current I3 OUTPUT B)

#	RANGO EBC (IUC RANGE)	VALOR APLICADO (APPLIED VALUE)	DESVÍO (DEVIATION)	INCERTIDUMBRE (UNCERTAINTY)
1	32,000 A	9,98103 A	0,190 A	1,7E-02 A

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NOTAS GENERALES. (General notes)

i. The deviation corresponds to the difference between the reading of the instrument under test and the reading of the reference instrument. It is considered positive when the reading of the former is greater than that of the latter.

ii. The measurement uncertainty is composed of the uncertainty of the calibration procedure and that of the instrument under calibration at the time the calibration is performed. Long-term components are not included. The expanded measurement uncertainty is reported as the standard measurement uncertainty multiplied by the coverage factor $k = 2$, which for a normal distribution corresponds to a coverage probability of approximately 95%. The standard measurement uncertainty has been determined in accordance with the publication "JCGM 100:2008 GUM 1995 with minor corrections. Evaluation of measurement data — Guide to the expression of uncertainty in measurement." The standard uncertainty has been determined according to the publication: "Guide to the expression of uncertainty in measurements" (2008) (BIMP, ISO, IEC, IFCC, IUPAC, IUPAP, OILM).

iii. This is to certify that the instrument whose calibration/test is detailed in this report was received in our calibration laboratory in good external physical condition, showing no visible alterations. Upon receipt, the calibration/test process was carried out in accordance with the corresponding internal procedure, taking into account the calibration points and the usage ranges provided by our client or by internal standards. The instrument is calibrated/tested in compliance with specifications, without any maintenance or adjustment prior to the process, thereby considering its initial conditions in their entirety.

iv. Conformity assessment: a calibrated value is considered "COMPLIANT" if the deviation plus the absolute value of the uncertainty is less than the tolerance. The tolerance is defined based on the specifications of the manufacturer of the instrument under calibration, unless the client has defined different tolerances.

v. The results contained in this document refer to the time and conditions under which the measurements were performed. INGENCA SRL is not responsible for any damages that may arise from the improper use of the instruments or the reports.

vi. This report may only be reproduced, in whole or in part, with the written authorization of INGENCA SRL.

vii. The reported results refer exclusively to the instrument under calibration (IUC).

NOTAS PARTICULARES. (Specific notes)